Brutal Practice Paper B46 - Weather & Climate, Decimals, Forests & Symmetry

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

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1. A hill station's midday temperatures for five days are 18, 16, 20, 15 and 21 degrees Celsius. The same week a nearby plain stays at 30 degrees Celsius at midday. How much cooler is the hill station's average midday temperature than the plain's? [Weather, Climate & Adaptations]

- **(A)** 9 degrees Celsius
- **(B)** 10 degrees Celsius
- **(C)** 48 degrees Celsius
- **(D)** 12 degrees Celsius
- **(E)** 18 degrees Celsius
- **(F)** 15 degrees Celsius

2. Work out the value of 3.6 + 0.45 - 1.8 + 2.05. [A Peek Beyond the Point]

- **(A)** 5.9
- **(B)** 2.25
- **(C)** 4.4
- **(D)** 4.2
- **(E)** 4.3
- **(F)** 3.3

3. The tallest trees in a forest form a canopy at 30 m. The understorey beneath reaches only 40 percent of that height. How tall is the understorey? [Forests]

- **(A)** 7.5 m
- **(B)** 18 m
- **(C)** 70 m
- **(D)** 30 m
- **(E)** 15 m
- **(F)** 12 m

4. How many lines of symmetry does a regular octagon (a regular eight-sided figure) have? [Symmetry]

- **(A)** 0
- **(B)** 2
- **(C)** 16
- **(D)** 4
- **(E)** 6
- **(F)** 8
- **(G)** 1

5. A town's total yearly rainfall is 1200 mm. The monsoon months bring 75 percent of it, and the rest falls in other months. How much rain falls outside the monsoon? [Weather, Climate & Adaptations]

- **(A)** 1200 mm
- **(B)** 75 mm
- **(C)** 900 mm
- **(D)** 400 mm
- **(E)** 300 mm
- **(F)** 150 mm
- **(G)** 25 mm

6. Multiply 2.5 by 0.04. [A Peek Beyond the Point]

- **(A)** 10
- **(B)** 0.01
- **(C)** 0.25
- **(D)** 0.001
- **(E)** 1.0
- **(F)** 0.4
- **(G)** 0.1

7. Each year a forest floor gains 5 kg of dead leaves per square metre, while decomposers break down 3 kg of it. By how much does the litter layer grow each year per square metre? [Forests]

- **(A)** 4 kg
- **(B)** 5 kg
- **(C)** 2 kg
- **(D)** 1.5 kg
- **(E)** 15 kg
- **(F)** 8 kg
- **(G)** 3 kg

8. A figure has rotational symmetry of order 5. What is the smallest angle, in degrees, through which it must be turned to look the same? [Symmetry]

- **(A)** 5 degrees
- **(B)** 360 degrees
- **(C)** 90 degrees
- **(D)** 36 degrees
- **(E)** 75 degrees
- **(F)** 60 degrees
- **(G)** 72 degrees

9. A migratory bird must fly 4500 km to a warmer region, covering 500 km each day. After flying steadily for 6 days, how much further must it still go? [Weather, Climate & Adaptations]

- **(A)** 1500 km
- **(B)** 3000 km
- **(C)** 1000 km
- **(D)** 750 km
- **(E)** 7500 km
- **(F)** 4500 km
- **(G)** 9 km

10. Divide 6.4 by 0.8. [A Peek Beyond the Point]

- **(A)** 8
- **(B)** 51.2
- **(C)** 0.8
- **(D)** 5.12
- **(E)** 0.125
- **(F)** 80
- **(G)** 0.08

11. One large tree releases about 120 kg of oxygen in a year. How much oxygen do 250 such trees release in a year? [Forests]

- **(A)** 30000 kg
- **(B)** 3000 kg
- **(C)** 300000 kg
- **(D)** 480 kg
- **(E)** 12000 kg
- **(F)** 370 kg

12. A shape looks exactly the same after being turned through 45 degrees. What is its order of rotational symmetry? [Symmetry]

- **(A)** 8
- **(B)** 6
- **(C)** 4
- **(D)** 45
- **(E)** 5
- **(F)** 9
- **(G)** 360

13. At dawn the temperature is -3 degrees Celsius. It then rises by 2 degrees Celsius every hour. What will the temperature be 6 hours after dawn? [Weather, Climate & Adaptations]

- **(A)** -15 degrees Celsius
- **(B)** -9 degrees Celsius
- **(C)** 3 degrees Celsius
- **(D)** 9 degrees Celsius
- **(E)** 15 degrees Celsius
- **(F)** 12 degrees Celsius

14. A pen costs Rs 12.50 and a notebook costs Rs 27.75. What change do you get from a Rs 100 note after buying one of each? [A Peek Beyond the Point]

- **(A)** Rs 60.25
- **(B)** Rs 72.25
- **(C)** Rs 59.75
- **(D)** Rs 59.25
- **(E)** Rs 40.75
- **(F)** Rs 59.50

15. Of all the rain falling on a forest, the dense canopy stops 30 percent from reaching the ground. If 80 mm of rain falls, how much actually reaches the forest floor? [Forests]

- **(A)** 80 mm
- **(B)** 26 mm
- **(C)** 50 mm
- **(D)** 30 mm
- **(E)** 110 mm
- **(F)** 56 mm
- **(G)** 24 mm

16. Among the capital letters H, A, N, O and Z, how many have at least one line of symmetry? [Symmetry]

- **(A)** 3
- **(B)** 2
- **(C)** 6
- **(D)** 0
- **(E)** 4
- **(F)** 1
- **(G)** 5

17. A polar bear weighs 400 kg, and 35 percent of its body weight is the fat that keeps it warm in the cold. How many kilograms of the bear is fat? [Weather, Climate & Adaptations]

- **(A)** 400 kg
- **(B)** 260 kg
- **(C)** 14 kg
- **(D)** 35 kg
- **(E)** 140 kg
- **(F)** 120 kg
- **(G)** 105 kg

18. What is 0.6 x 0.6 x 0.6? [A Peek Beyond the Point]

- **(A)** 0.216
- **(B)** 1.8
- **(C)** 0.0216
- **(D)** 0.36
- **(E)** 0.06
- **(F)** 2.16

19. Only about one-tenth of the energy passes from one level of a food chain to the next. If grass holds 8000 units of energy, how much energy reaches the level that eats the plant-eaters? [Forests]

- **(A)** 80 units
- **(B)** 0.8 units
- **(C)** 160 units
- **(D)** 8 units
- **(E)** 800 units
- **(F)** 8000 units
- **(G)** 240 units

20. Take an equilateral triangle. Add its count of mirror (symmetry) lines to its order of rotational symmetry. What total do you get? [Symmetry]

- **(A)** 4
- **(B)** 6
- **(C)** 0
- **(D)** 5
- **(E)** 12
- **(F)** 9
- **(G)** 3

21. A city's yearly rainfall over four years was 800, 950, 700 and 950 mm. What average yearly rainfall best describes its climate? [Weather, Climate & Adaptations]

- **(A)** 950 mm
- **(B)** 800 mm
- **(C)** 850 mm
- **(D)** 3400 mm
- **(E)** 900 mm
- **(F)** 750 mm
- **(G)** 825 mm

22. In the number 47.063, what is the place value of the digit 6? [A Peek Beyond the Point]

- **(A)** 0.063
- **(B)** 6
- **(C)** 0.63
- **(D)** 0.6
- **(E)** 0.006
- **(F)** 0.06
- **(G)** 60

23. A forest once covered 500 hectares. After clearing, only 350 hectares remain. What percentage of the forest was lost? [Forests]

- **(A)** 43 percent
- **(B)** 70 percent
- **(C)** 30 percent
- **(D)** 150 percent
- **(E)** 35 percent
- **(F)** 60 percent
- **(G)** 15 percent

24. Consider a rectangle whose length and breadth are different (so it is not a square). Count its mirror lines of symmetry. [Symmetry]

- **(A)** 1
- **(B)** 3
- **(C)** 4
- **(D)** 8
- **(E)** 2
- **(F)** 6

25. A camel can lose up to 25 percent of its body water and still survive. If a camel's body holds 80 litres of water, how many litres can it lose and still live? [Weather, Climate & Adaptations]

- **(A)** 25 litres
- **(B)** 80 litres
- **(C)** 16 litres
- **(D)** 60 litres
- **(E)** 20 litres
- **(F)** 40 litres
- **(G)** 5 litres

26. Write 3/4 and 2/5 as decimals and add them together. [A Peek Beyond the Point]

- **(A)** 11.5
- **(B)** 1.2
- **(C)** 0.115
- **(D)** 1.15
- **(E)** 1.5
- **(F)** 1.05
- **(G)** 0.79

27. One tree drops about 2000 seeds in a year, but only 2 percent of them grow into saplings. How many saplings come from one tree in a year? [Forests]

- **(A)** 400 saplings
- **(B)** 100 saplings
- **(C)** 40 saplings
- **(D)** 20 saplings
- **(E)** 200 saplings
- **(F)** 2000 saplings
- **(G)** 4 saplings

28. A square is turned about its centre through one full turn. In how many different positions (counting the starting one) does it look exactly the same? [Symmetry]

- **(A)** 1
- **(B)** 3
- **(C)** 8
- **(D)** 2
- **(E)** 360
- **(F)** 4
- **(G)** 90

29. Temperature falls about 6 degrees Celsius for every 1000 m climbed up a mountain. At the base it is 28 degrees Celsius. What is the temperature 2500 m higher up? [Weather, Climate & Adaptations]

- **(A)** 13 degrees Celsius
- **(B)** 18 degrees Celsius
- **(C)** 15 degrees Celsius
- **(D)** 43 degrees Celsius
- **(E)** 16 degrees Celsius
- **(F)** 12 degrees Celsius
- **(G)** 10 degrees Celsius

30. Find the average of 2.4, 3.6 and 5.1. [A Peek Beyond the Point]

(A) 5.55
(B) 4.0
(C) 11.1
(D) 3.5
(E) 3.8
(F) 3.7
(G) 3.6

31. A tree's roots hold soil and store 0.25 cubic metres of water each. How much water do the roots of 60 such trees hold together? [Forests]

(A) 1.5 cubic metres
(B) 24 cubic metres
(C) 240 cubic metres
(D) 15 cubic metres
(E) 150 cubic metres
(F) 30 cubic metres
(G) 60.25 cubic metres

32. Through what smallest angle can a regular hexagon be rotated so that it looks unchanged? [Symmetry]

(A) 360 degrees
(B) 60 degrees
(C) 90 degrees
(D) 6 degrees
(E) 30 degrees
(F) 72 degrees
(G) 120 degrees

33. Rain of 20 mm falls on a flat roof measuring 5 m by 4 m. Since 1 mm of rain on 1 square metre gives 1 litre of water, how many litres fall on the roof? [Weather, Climate & Adaptations]

(A) 100 litres
(B) 200 litres
(C) 400 litres
(D) 800 litres
(E) 4000 litres
(F) 20 litres
(G) 40 litres

34. Subtract 4.875 from 9.2. [A Peek Beyond the Point]

(A) 4.375
(B) 5.325
(C) 5.475
(D) 4.435
(E) 4.325
(F) 13.075

35. A forest absorbs about 6 kg of carbon dioxide for every 1 kg of wood it grows. To grow 45 kg of wood, how much carbon dioxide does it absorb? [Forests]

- **(A)** 2700 kg
- **(B)** 90 kg
- **(C)** 270 kg
- **(D)** 51 kg
- **(E)** 45 kg
- **(F)** 7.5 kg
- **(G)** 135 kg

36. A regular polygon has 12 lines of symmetry. Through what smallest angle does it map onto itself when rotated? [Symmetry]

- **(A)** 12 degrees
- **(B)** 45 degrees
- **(C)** 24 degrees
- **(D)** 60 degrees
- **(E)** 360 degrees
- **(F)** 15 degrees
- **(G)** 30 degrees

37. In summer a place gets 15 hours of daylight each day; in winter only 9 hours. Over a 30-day summer month, how many more daylight hours are there in total than in a 30-day winter month? [Weather, Climate & Adaptations]

- **(A)** 180 hours
- **(B)** 90 hours
- **(C)** 720 hours
- **(D)** 6 hours
- **(E)** 450 hours
- **(F)** 360 hours
- **(G)** 24 hours

38. A rectangular tile measures 0.5 m by 0.3 m. What is its area in square metres? [A Peek Beyond the Point]

- **(A)** 1.5
- **(B)** 0.53
- **(C)** 0.16
- **(D)** 0.30
- **(E)** 0.015
- **(F)** 0.15

39. A patch of forest has 12 kinds of trees, and each kind of tree supports about 15 kinds of insects. About how many kinds of insects live in the patch? [Forests]

- **(A)** 180
- **(B)** 175
- **(C)** 120
- **(D)** 200
- **(E)** 150
- **(F)** 27
- **(G)** 160

40. A parallelogram has rotational symmetry of order 2 but no line of symmetry. Through what smallest angle does it look the same? [Symmetry]

- **(A)** 360 degrees
- **(B)** 2 degrees
- **(C)** 90 degrees
- **(D)** 180 degrees
- **(E)** 45 degrees
- **(F)** 270 degrees

41. Snow falls steadily at 2.5 cm per hour. The ground already has 4 cm of snow on it. How deep is the snow after 6 more hours of snowfall? [Weather, Climate & Adaptations]

- **(A)** 19 cm
- **(B)** 25 cm
- **(C)** 6.5 cm
- **(D)** 15 cm
- **(E)** 12.5 cm
- **(F)** 16.5 cm

42. Petrol costs Rs 102.50 per litre. What do 4 litres cost? [A Peek Beyond the Point]

- **(A)** Rs 412
- **(B)** Rs 41
- **(C)** Rs 4100
- **(D)** Rs 406.50
- **(E)** Rs 408
- **(F)** Rs 410

43. Of 240 kg of fallen leaves, three-eighths turns into dark, rich humus. How many kilograms of humus form? [Forests]

- **(A)** 30 kg
- **(B)** 45 kg
- **(C)** 150 kg
- **(D)** 64 kg
- **(E)** 80 kg
- **(F)** 90 kg
- **(G)** 120 kg

44. Draw a regular five-sided figure next to a regular six-sided figure. Adding up each one's mirror lines, how many lines of symmetry are there in all? [Symmetry]

- **(A)** 10
- **(B)** 11
- **(C)** 30
- **(D)** 12
- **(E)** 9
- **(F)** 1

45. Air at 30 degrees Celsius can hold at most 30 grams of water vapour per cubic metre. If the air actually holds 18 grams per cubic metre, what is the relative humidity (the actual amount as a percent of the maximum)? [Weather, Climate & Adaptations]

- **(A)** 167 percent
- **(B)** 60 percent
- **(C)** 50 percent
- **(D)** 48 percent
- **(E)** 30 percent
- **(F)** 18 percent
- **(G)** 12 percent

46. Write the decimal 0.36 as a fraction in its lowest terms. [A Peek Beyond the Point]

- **(A)** 36/100
- **(B)** 9/20
- **(C)** 18/50
- **(D)** 4/25
- **(E)** 9/25
- **(F)** 3/25
- **(G)** 36/10

47. A tree adds one growth ring each year and grows 0.4 cm thicker in radius every year. After 25 years, by how much has its radius increased? [Forests]

- **(A)** 10 cm
- **(B)** 12.5 cm
- **(C)** 25.4 cm
- **(D)** 4 cm
- **(E)** 62.5 cm
- **(F)** 100 cm
- **(G)** 1 cm

48. A windmill design has 4 identical blades equally spaced around the centre. Through what smallest angle can it turn to look the same? [Symmetry]

- **(A)** 90 degrees
- **(B)** 180 degrees
- **(C)** 60 degrees
- **(D)** 120 degrees
- **(E)** 360 degrees
- **(F)** 45 degrees

49. Four winter nights record temperatures of -5, -2, -8 and -1 degrees Celsius. What is the average night temperature? [Weather, Climate & Adaptations]

- **(A)** -4 degrees Celsius
- **(B)** 0 degrees Celsius
- **(C)** -3.5 degrees Celsius
- **(D)** -16 degrees Celsius
- **(E)** -5 degrees Celsius
- **(F)** 4 degrees Celsius
- **(G)** -8 degrees Celsius

50. Rice costs Rs 48.40 per kilogram. You buy 2.5 kilograms. How much do you pay? [A Peek Beyond the Point]

- **(A)** Rs 12.10
- **(B)** Rs 121.50
- **(C)** Rs 121
- **(D)** Rs 120.40
- **(E)** Rs 484
- **(F)** Rs 96.80
- **(G)** Rs 122

51. A forest fire spreads outward at 3 m per minute. How far does its edge move in 2 hours? [Forests]

- **(A)** 360 m
- **(B)** 36 m
- **(C)** 3600 m
- **(D)** 720 m
- **(E)** 6 m
- **(F)** 180 m

52. A shape is turned 90 degrees at a time about its centre. How many of these 90-degree turns are needed to bring it back to its exact starting position? [Symmetry]

- **(A)** 3
- **(B)** 8
- **(C)** 2
- **(D)** 1
- **(E)** 90
- **(F)** 4
- **(G)** 360

53. A seal's blubber is 6 cm thick and slows heat loss. A whale's blubber is 2.5 times as thick. How thick is the whale's blubber? [Weather, Climate & Adaptations]

- **(A)** 30 cm
- **(B)** 3.5 cm
- **(C)** 15 cm
- **(D)** 2.5 cm
- **(E)** 12 cm
- **(F)** 8.5 cm

54. Round 7.486 to two decimal places. [A Peek Beyond the Point]

- **(A)** 8
- **(B)** 7.49
- **(C)** 7.50
- **(D)** 7.48
- **(E)** 7.4
- **(F)** 7.5
- **(G)** 7.486

55. Four forest patches receive 1200, 1500, 900 and 1600 mm of rain a year. What is the average yearly rainfall across the patches? [Forests]

- **(A)** 1500 mm
- **(B)** 1400 mm
- **(C)** 1250 mm
- **(D)** 1300 mm
- **(E)** 5200 mm
- **(F)** 1200 mm
- **(G)** 1350 mm

56. Among the capital letters S, T, X and O, how many have rotational symmetry of order 2 (they look the same after a half turn)? [Symmetry]

- **(A)** 6
- **(B)** 4
- **(C)** 5
- **(D)** 2
- **(E)** 3
- **(F)** 1

57. Near the equator the Sun is almost overhead all year, so a region gets about 12 hours of daylight every single day. How many hours of daylight does it get in a 31-day month? [Weather, Climate & Adaptations]

- **(A)** 43 hours
- **(B)** 372 hours
- **(C)** 412 hours
- **(D)** 31 hours
- **(E)** 360 hours
- **(F)** 248 hours
- **(G)** 144 hours

58. A ribbon 4.5 m long is cut into equal pieces, each 0.75 m long. How many pieces are made? [A Peek Beyond the Point]

- **(A)** 0.6
- **(B)** 60
- **(C)** 5
- **(D)** 3.375
- **(E)** 6
- **(F)** 0.166
- **(G)** 6.5

59. A tree photosynthesises only during the 10 daylight hours of a 24-hour day. What fraction of the day, in lowest terms, does it spend photosynthesising? [Forests]

- **(A)** 5/24
- **(B)** 10/24
- **(C)** 5/6
- **(D)** 5/12
- **(E)** 1/2
- **(F)** 2/5

60. A clock shows the time 3:00. When seen in a mirror, what time does the clock appear to show? [Symmetry]

- **(A)** 3:30
- **(B)** 3:00
- **(C)** 8:00
- **(D)** 9:30
- **(E)** 9:00
- **(F)** 12:00

61. At sea level a climber's heart beats 72 times a minute. High on a mountain, where the air is thin, it beats 25 percent faster. What is the high-altitude heart rate? [Weather, Climate & Adaptations]

- **(A)** 96 beats per minute
- **(B)** 108 beats per minute
- **(C)** 18 beats per minute
- **(D)** 54 beats per minute
- **(E)** 75 beats per minute
- **(F)** 90 beats per minute

62. Three items cost Rs 15.60, Rs 9.45 and Rs 24.95. What is the total cost? [A Peek Beyond the Point]

- **(A)** Rs 48.90
- **(B)** Rs 50.05
- **(C)** Rs 51
- **(D)** Rs 49
- **(E)** Rs 50.10
- **(F)** Rs 50
- **(G)** Rs 49.95

63. In one gram of forest soil there are about 4 x 10^7 bacteria. Using powers of ten, how many bacteria are in 5 grams of the soil? [Forests]

- **(A)** 4 x 10^8
- **(B)** 45 x 10^6
- **(C)** 2 x 10^8
- **(D)** 2 x 10^7
- **(E)** 2 x 10^6
- **(F)** 8 x 10^7
- **(G)** 2 x 10^9

64. Picture a triangle with exactly two equal sides (an isosceles triangle that is not equilateral). How many mirror lines of symmetry can you draw on it? [Symmetry]

- **(A)** 6
- **(B)** 3
- **(C)** 2
- **(D)** 0
- **(E)** 1
- **(F)** 4
- **(G)** 5

65. Monthly rainfall in mm for four months is 120, 80, 200 and 160. The driest month's rainfall is what fraction of the wettest month's, in lowest terms? [Weather, Climate & Adaptations]

(A) 5/2
(B) 3/5
(C) 80/160
(D) 1/2
(E) 2/3
(F) 4/5
(G) 2/5

66. Which of these is the largest number: 0.7, 0.68, 0.709 or 0.71? [A Peek Beyond the Point]

(A) 0.71
(B) 0.078
(C) 0.68
(D) 0.067
(E) 0.701
(F) 0.709
(G) 0.7

67. Without trees, a slope loses 8 cm of topsoil in 10 years. With a forest cover, it loses only 1 cm in the same time. How much soil does the forest save over 10 years? [Forests]

(A) 7.5 cm
(B) 0.7 cm
(C) 80 cm
(D) 8 cm
(E) 9 cm
(F) 1 cm
(G) 7 cm

68. Think about a perfectly round circle. Counting every possible mirror line you could fold it along, how many lines of symmetry does it have? [Symmetry]

(A) 8
(B) 4
(C) 1
(D) 2
(E) infinitely many
(F) 0
(G) 360

69. Over 100 years a region's average temperature rose from 24.6 degrees Celsius to 26.1 degrees Celsius. What was the total rise? [Weather, Climate & Adaptations]

(A) 50.7 degrees Celsius
(B) 0.5 degrees Celsius
(C) 2.1 degrees Celsius
(D) 1.4 degrees Celsius
(E) 1.6 degrees Celsius
(F) 1.5 degrees Celsius

70. What is 1.2 multiplied by itself (1.2 squared)? [A Peek Beyond the Point]

- **(A)** 14.4
- **(B)** 1.4
- **(C)** 0.144
- **(D)** 1.44
- **(E)** 2.4
- **(F)** 2.44

71. A single banyan's spreading canopy shades the ground. Treating the shaded patch as a square of side 20 m, what ground area does it shade? [Forests]

- **(A)** 200 square metres
- **(B)** 100 square metres
- **(C)** 800 square metres
- **(D)** 4000 square metres
- **(E)** 40 square metres
- **(F)** 80 square metres
- **(G)** 400 square metres

72. A regular polygon maps onto itself every time it is turned 40 degrees. How many sides does it have? [Symmetry]

- **(A)** 320 sides
- **(B)** 10 sides
- **(C)** 7 sides
- **(D)** 9 sides
- **(E)** 40 sides
- **(F)** 8 sides
- **(G)** 36 sides

73. Huddling together lets penguins save 50 percent of the heat they would otherwise lose. If a lone penguin loses 240 units of heat per hour, how much does a huddling penguin lose? [Weather, Climate & Adaptations]

- **(A)** 190 units
- **(B)** 200 units
- **(C)** 60 units
- **(D)** 120 units
- **(E)** 480 units
- **(F)** 100 units
- **(G)** 240 units

74. A wire is 2.5 m long. Express its length in centimetres. [A Peek Beyond the Point]

- **(A)** 25 cm
- **(B)** 250 cm
- **(C)** 2500 cm
- **(D)** 205 cm
- **(E)** 0.025 cm
- **(F)** 25.0 cm

75. It takes about 17 trees to make 1 tonne of paper. How many trees are needed to make 6 tonnes of paper? [Forests]

- **(A)** 1020 trees
- **(B)** 23 trees
- **(C)** 120 trees
- **(D)** 6 trees
- **(E)** 102 trees
- **(F)** 11 trees
- **(G)** 96 trees

76. A point lies 5 units to the right of a vertical mirror line. Where is its mirror image? [Symmetry]

- **(A)** 10 units to the left of the line
- **(B)** 5 units to the right of the line
- **(C)** 2.5 units to the left of the line
- **(D)** 5 units to the left of the line
- **(E)** 5 units above the line
- **(F)** on the line itself
- **(G)** 10 units to the right of the line

77. A bird migrates 3200 km south in autumn and the same distance back north in spring. If it does this every year, how far does it migrate altogether in 3 years? [Weather, Climate & Adaptations]

- **(A)** 12800 km
- **(B)** 9600 km
- **(C)** 6400 km
- **(D)** 3200 km
- **(E)** 16000 km
- **(F)** 19200 km

78. You buy 3 pencils at Rs 4.50 each and pay with a Rs 20 note. How much change should you get? [A Peek Beyond the Point]

- **(A)** Rs 5.50
- **(B)** Rs 4.50
- **(C)** Rs 7.50
- **(D)** Rs 6.50
- **(E)** Rs 6
- **(F)** Rs 16.50

79. In a food chain there are 1000 grass plants, one-tenth as many deer as grass plants, and one-fifth as many tigers as deer. How many tigers are there? [Forests]

- **(A)** 100 tigers
- **(B)** 50 tigers
- **(C)** 500 tigers
- **(D)** 10 tigers
- **(E)** 200 tigers
- **(F)** 2 tigers
- **(G)** 20 tigers

80. A figure has 6 lines of symmetry and rotational symmetry of order 6. If you multiply the number of lines by the order, what number do you get? [Symmetry]

- **(A)** 36
- **(B)** 6
- **(C)** 60
- **(D)** 1
- **(E)** 12
- **(F)** 360
- **(G)** 30

81. A place has a rainy season that lasts 4 months of the year. What fraction of the whole year is the rainy season, in lowest terms? [Weather, Climate & Adaptations]

- **(A)** 2/3
- **(B)** 1/3
- **(C)** 3/4
- **(D)** 1/2
- **(E)** 4/8
- **(F)** 2/5

82. Add 0.9, 0.09 and 0.009. [A Peek Beyond the Point]

- **(A)** 9.99
- **(B)** 0.99
- **(C)** 0.099
- **(D)** 0.27
- **(E)** 2.7
- **(F)** 0.999

83. A big tree releases about 250 litres of water vapour into the air each day. How many litres does it release during the whole month of April, which has 30 days? [Forests]

- **(A)** 750 litres
- **(B)** 7500 litres
- **(C)** 280 litres
- **(D)** 3000 litres
- **(E)** 75000 litres
- **(F)** 6000 litres

84. A shape has a vertical line of symmetry. Its left half has an area of 18 square cm. What is the total area of the whole shape? [Symmetry]

- **(A)** 36 square cm
- **(B)** 18 square cm
- **(C)** 27 square cm
- **(D)** 20 square cm
- **(E)** 72 square cm
- **(F)** 360 square cm

85. Fresh snow reflects about 80 percent of the sunlight that hits it. If 250 units of sunlight reach the snow, how many units are absorbed rather than reflected? [Weather, Climate & Adaptations]

- **(A)** 200 units
- **(B)** 50 units
- **(C)** 170 units
- **(D)** 20 units
- **(E)** 100 units
- **(F)** 30 units
- **(G)** 80 units

86. A car covers 0.6 km in 0.5 minutes at a steady speed. At the same speed, how far does it travel in 2 minutes? [A Peek Beyond the Point]

- **(A)** 2.4 km
- **(B)** 1.2 km
- **(C)** 0.6 km
- **(D)** 1.1 km
- **(E)** 2.0 km
- **(F)** 3.0 km
- **(G)** 4.8 km

87. Volunteers plant 45 saplings every day. How many days will it take them to plant 1350 saplings? [Forests]

- **(A)** 60750 days
- **(B)** 30 days
- **(C)** 3 days
- **(D)** 300 days
- **(E)** 45 days
- **(F)** 25 days

88. In a regular hexagon, the 6 lines of symmetry all pass through the centre. What is the angle between two neighbouring lines of symmetry? [Symmetry]

- **(A)** 6 degrees
- **(B)** 15 degrees
- **(C)** 60 degrees
- **(D)** 30 degrees
- **(E)** 360 degrees
- **(F)** 36 degrees

89. A desert day reaches 45.5 degrees Celsius, and the following night drops to 8.5 degrees Celsius. By how much does the temperature fall from day to night? [Weather, Climate & Adaptations]

- **(A)** 53 degrees Celsius
- **(B)** 37 degrees Celsius
- **(C)** 36 degrees Celsius
- **(D)** 38 degrees Celsius
- **(E)** 37.5 degrees Celsius
- **(F)** 54 degrees Celsius

90. Find 0.1 of 0.1. [A Peek Beyond the Point]

- **(A)** 0.01
- **(B)** 1.0
- **(C)** 0.2
- **(D)** 0.02
- **(E)** 0.11
- **(F)** 0.1

91. A forest yields 800 kg of wild fruit, and animals eat 65 percent of it. How much fruit is left? [Forests]

- **(A)** 300 kg
- **(B)** 735 kg
- **(C)** 260 kg
- **(D)** 480 kg
- **(E)** 520 kg
- **(F)** 65 kg
- **(G)** 280 kg

92. A regular polygon has an angle of rotational symmetry of 24 degrees. How many lines of symmetry does it have? [Symmetry]

- **(A)** 24
- **(B)** 10
- **(C)** 20
- **(D)** 336
- **(E)** 30
- **(F)** 15
- **(G)** 12

93. The average rainfall over 3 months is 90 mm. Two of the months had 80 mm and 110 mm. How much rain fell in the third month? [Weather, Climate & Adaptations]

- **(A)** 90 mm
- **(B)** 60 mm
- **(C)** 95 mm
- **(D)** 190 mm
- **(E)** 100 mm
- **(F)** 80 mm
- **(G)** 70 mm

94. A bill of Rs 90.60 is shared equally among 3 friends. How much does each friend pay? [A Peek Beyond the Point]

- **(A)** Rs 30.02
- **(B)** Rs 30.20
- **(C)** Rs 31.20
- **(D)** Rs 30.60
- **(E)** Rs 271.80
- **(F)** Rs 27.18
- **(G)** Rs 30

95. A tree casts a 12 m shadow at the same moment a 2 m pole casts a 1.5 m shadow. How tall is the tree? [Forests]

- **(A)** 24 m
- **(B)** 9 m
- **(C)** 14 m
- **(D)** 18 m
- **(E)** 16 m
- **(F)** 12 m

96. An equilateral triangle is rotated about its centre. Not counting the start (0 degrees) or a full turn (360 degrees), at how many angles does it look exactly the same? [Symmetry]

- **(A)** 3
- **(B)** 0
- **(C)** 6
- **(D)** 1
- **(E)** 2
- **(F)** 4
- **(G)** 120

97. A glacier's length shrank from 50 km to 40 km over many years. What percentage of its length did it lose? [Weather, Climate & Adaptations]

- **(A)** 80 percent
- **(B)** 20 percent
- **(C)** 40 percent
- **(D)** 15 percent
- **(E)** 10 percent
- **(F)** 12.5 percent

98. A number multiplied by 0.5 gives 3.6. What is the number? [A Peek Beyond the Point]

- **(A)** 0.72
- **(B)** 1.8
- **(C)** 18
- **(D)** 3.1
- **(E)** 7.1
- **(F)** 4.1
- **(G)** 7.2

99. Each forest layer lets only half the sunlight pass through to the layer below. If 1600 units of sunlight hit the canopy, how much reaches the forest floor after passing through 3 layers? [Forests]

- **(A)** 100 units
- **(B)** 200 units
- **(C)** 800 units
- **(D)** 160 units
- **(E)** 533 units
- **(F)** 50 units

100. A figure is made of a square with an equilateral triangle sitting on its top side, like a simple house outline. How many lines of symmetry does the whole house shape have? [Symmetry]

- **(A)** 1
- **(B)** 7
- **(C)** 2
- **(D)** 3
- **(E)** 0
- **(F)** 4